

R Lab 1

Use R to perform analyses on two datasets. Use the R Cookbook to figure out how to do all the analyses.

Datasets to analyze (R built in datasets)

- (1) airquality
- (2) ToothGrowth

1. Show the top of the dataset using head
2. Attach variables to be easier to analyze: attach(dataframe)
3. Use 3 different methods to determine normality: qqnorm, a histogram, stats test
4. Figure out which transformation could make it normal
5. Transform variables as necessary
6. Make scatterplots of all the non-categorical variables
7. Test all pairs of correlations
8. Adjust p-value using fdr and Bonferroni

Specific tests

ToothGrowth:

Perform an ANOVA with supp as the factor.

Perform a linear regression of len with dose

Airquality:

Perform an ANOVA of temp with month as factor

Correlate Ozone with Solar.R, Wind with Temp